

THE NEUTRINOS EMISSION FROM SN EXPLOSION



Weekly Analysis Meeting – 20/03/25

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WHAT HAS BEEN DONE?

Investigated physical scenario;

Studied the emission model;

Produced the sources catalog;

Plan for different analysis strategies.

CUT&COUNT ANALYSIS

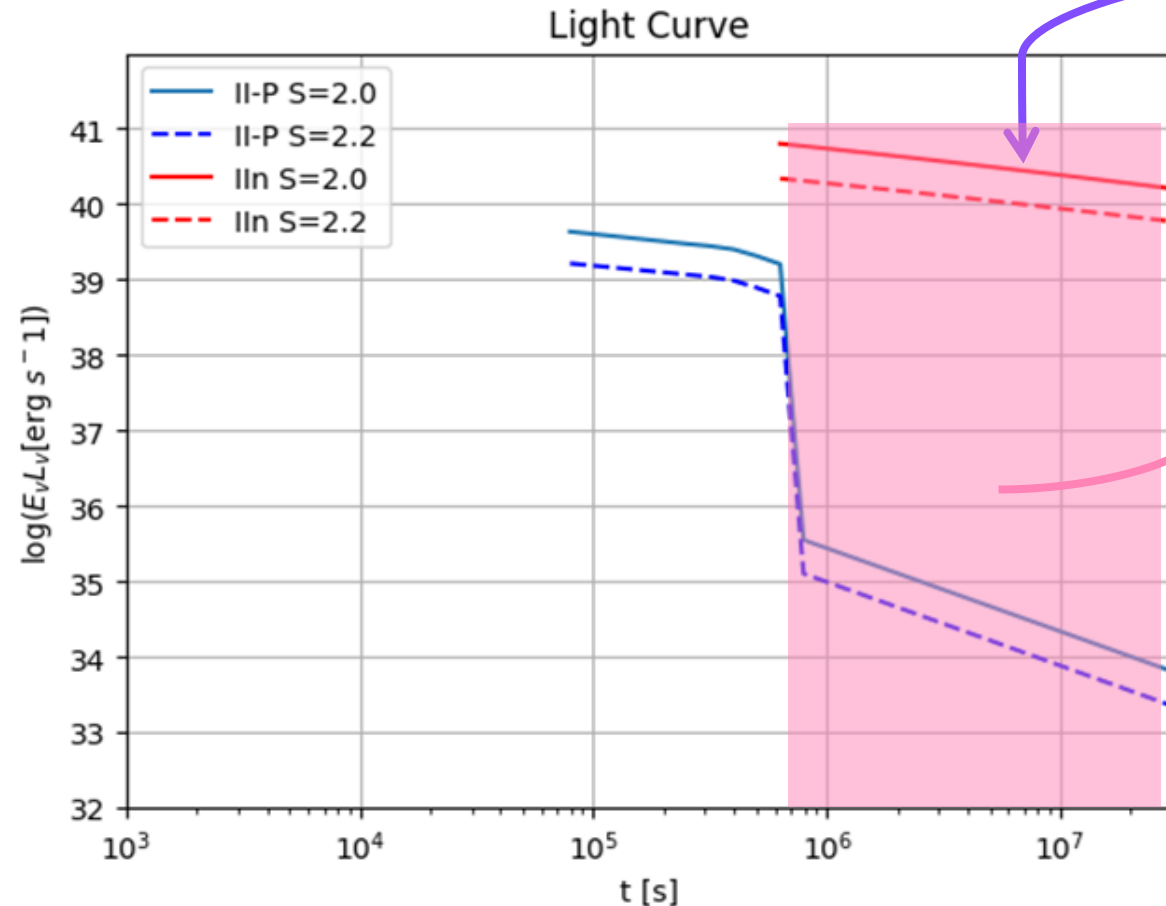
Starting with MRF calculation -> template from the collaboration -> Calcrate2.py in Aanet

BUT Calcrate code is not used for time dependent analysis

What did we do?

1. Using my code to compute the number of signal events from the Murase's model;
2. Using Calcrate2.py to compute the number of background events;
3. Check if the results are in agreement using GraphFlux() function.

CUT&COUNT ANALYSIS



We studied first SN IIn emission model, with $s=2.0$.

We have analyzed the time window between $t_{\text{onset}} = 10^{5.8}$ and $t_{\text{vmax}} = 10^{7.5}$

$$t_{\text{vmax}} - t_{\text{onset}} = 358,7 \text{ days} \sim 1 \text{ years}$$

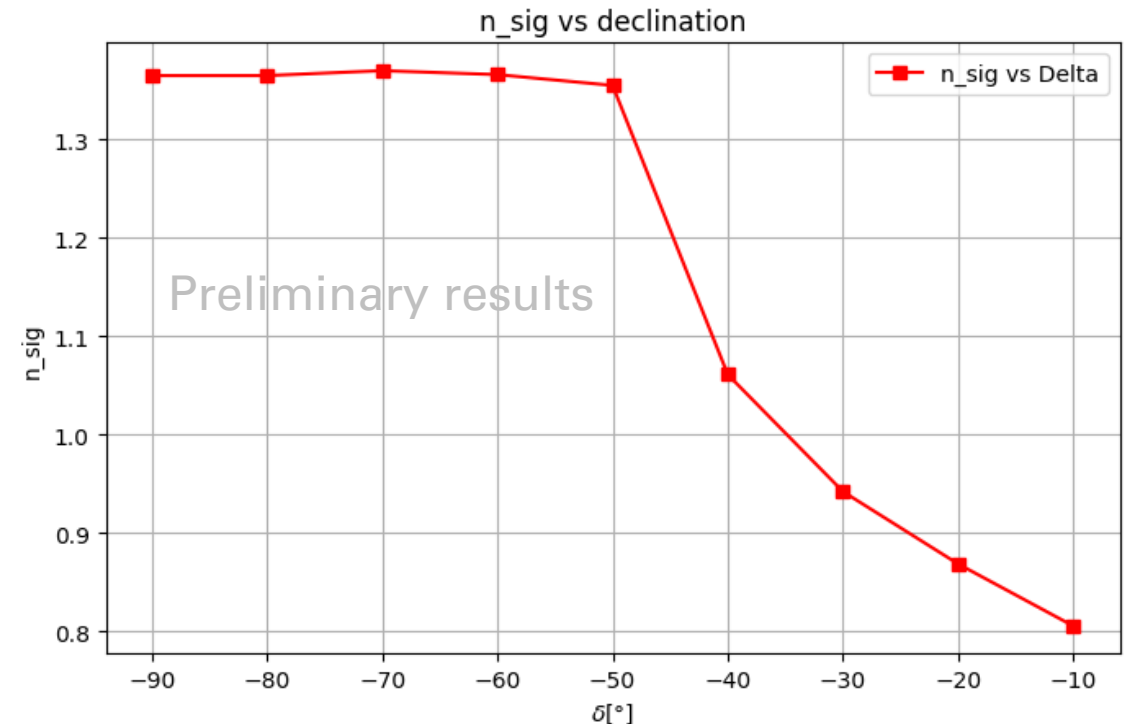
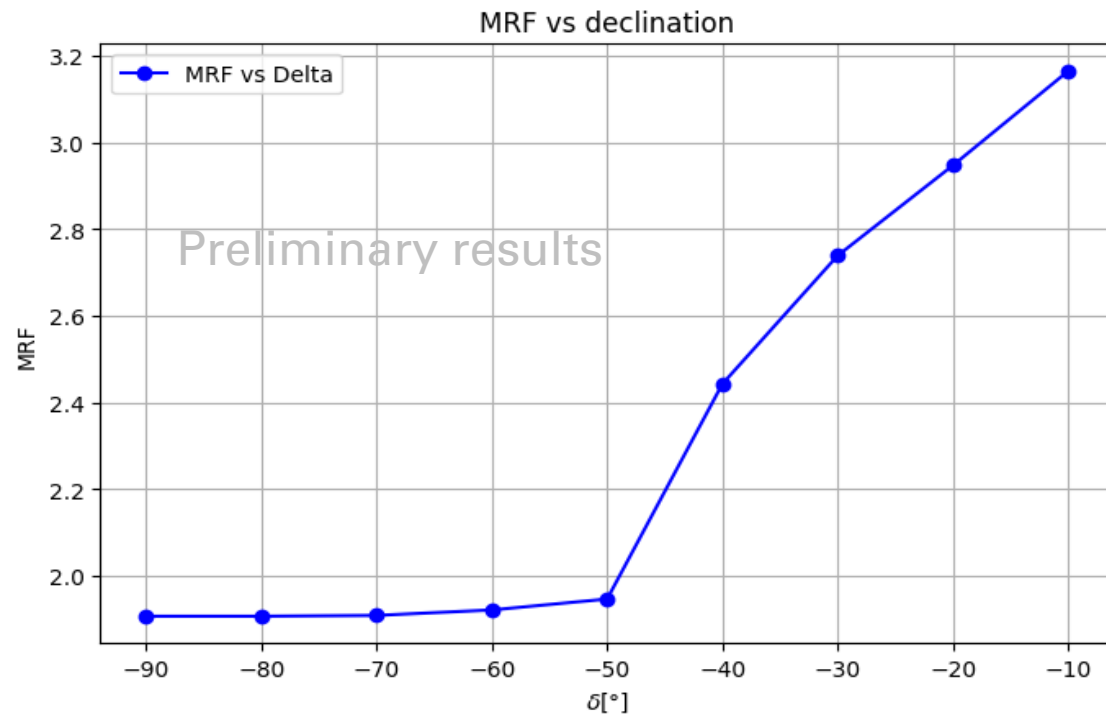
MRF – my code (ν_μ)

Detector configuration: ARCA21

(202475_5_53_29_DETECTORESPONSE_ARCA21_133_tmva_tracks_v10.ALL_zen_b40_pre_upm01_antinoise_bdt095.root)

Source distance: 1Mpc

Supernova Model: SN IIn with CSM interactions



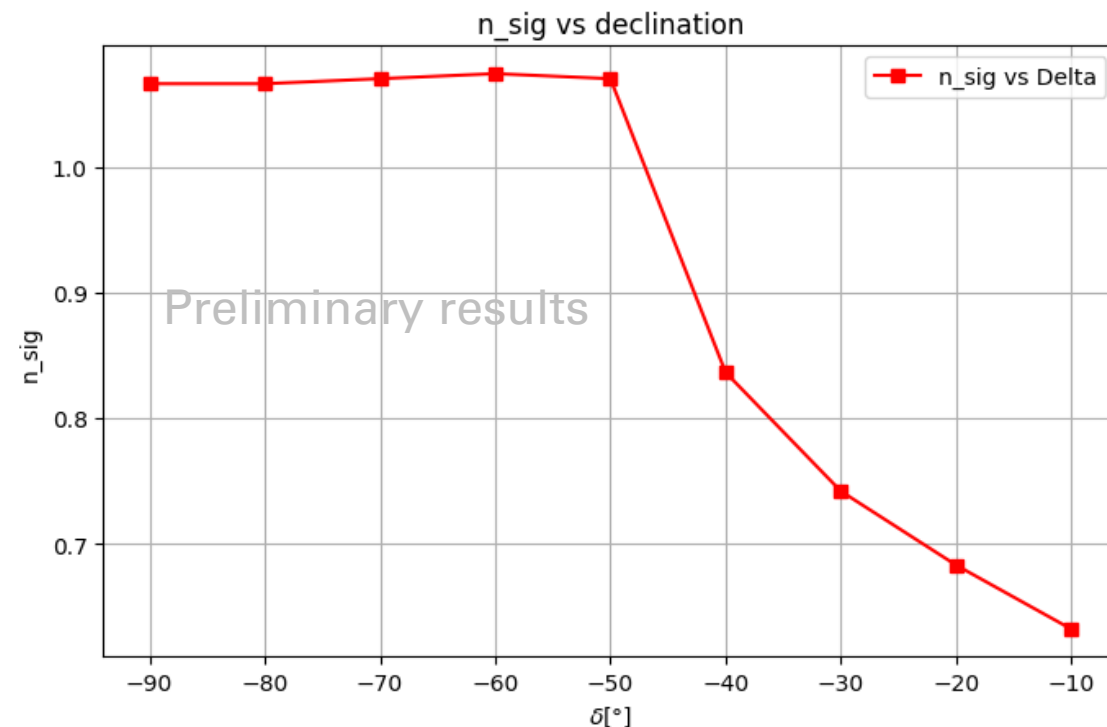
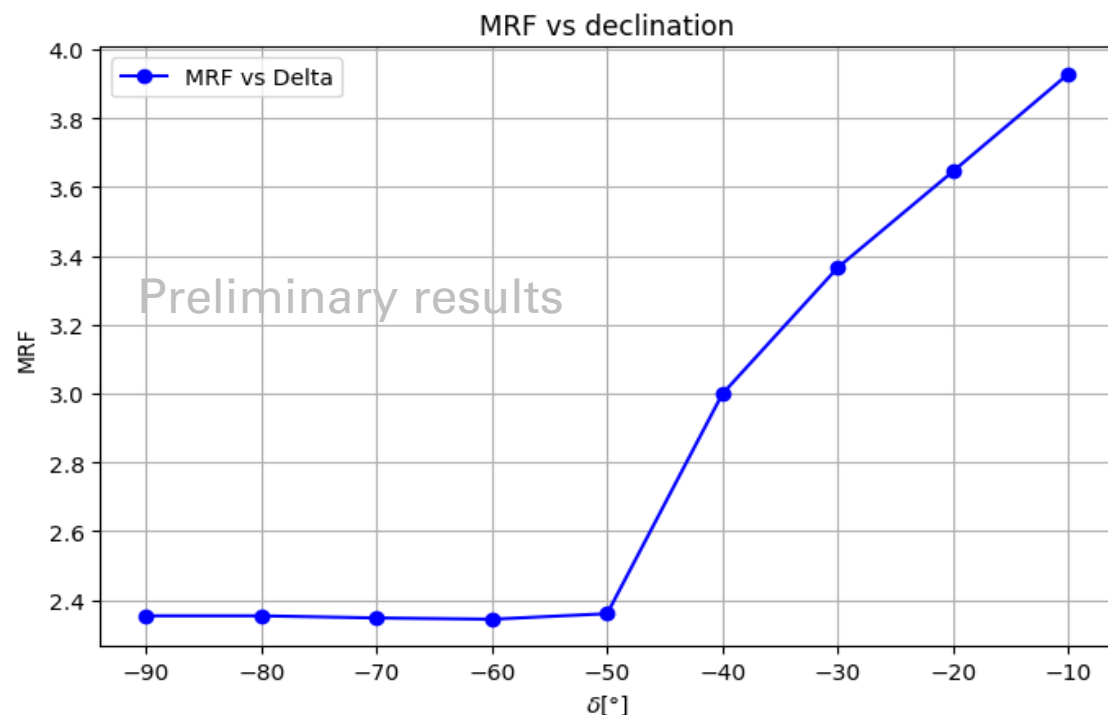
MRF – my code ($\overline{\nu}_\mu$)

Detector configuration: ARCA21

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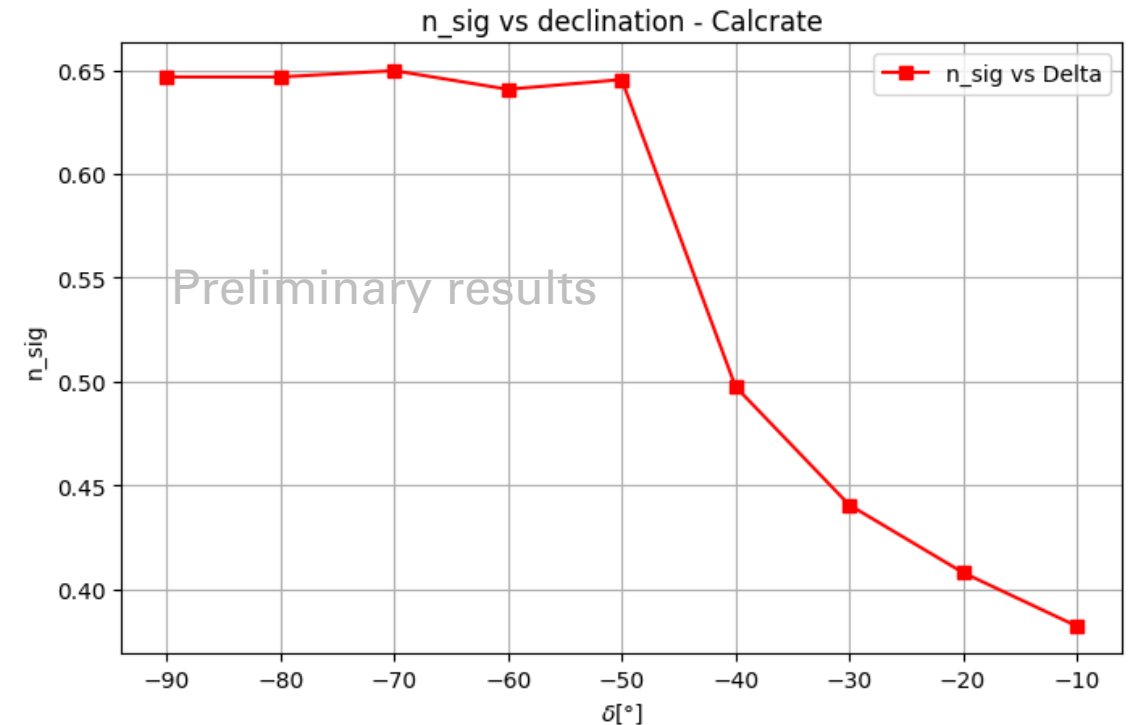
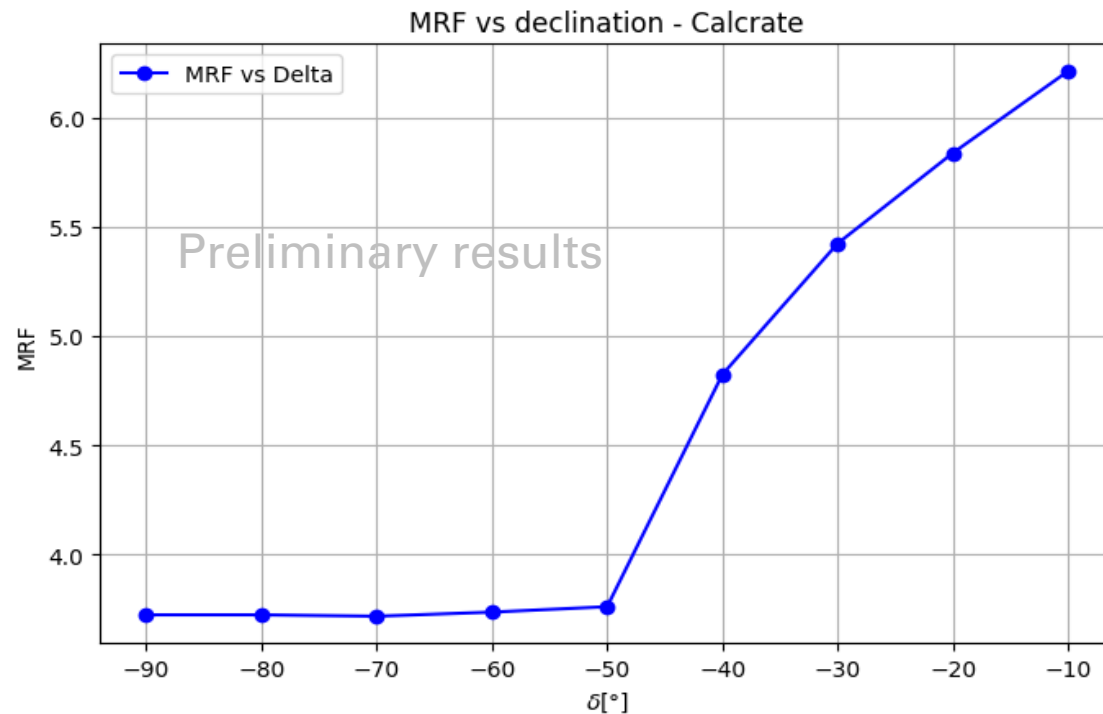
MRF – Calcrate (ν_μ)

Detector configuration: ARCA21

(202475_5_53_29_DETECTORESPONSE_ARCA21_133_tmva_tracks_v10.ALL_zen_b40_pre_upm01_antinoise_bdt095.root)

Source distance: 1Mpc

Supernova Model: SN IIn with CSM interactions



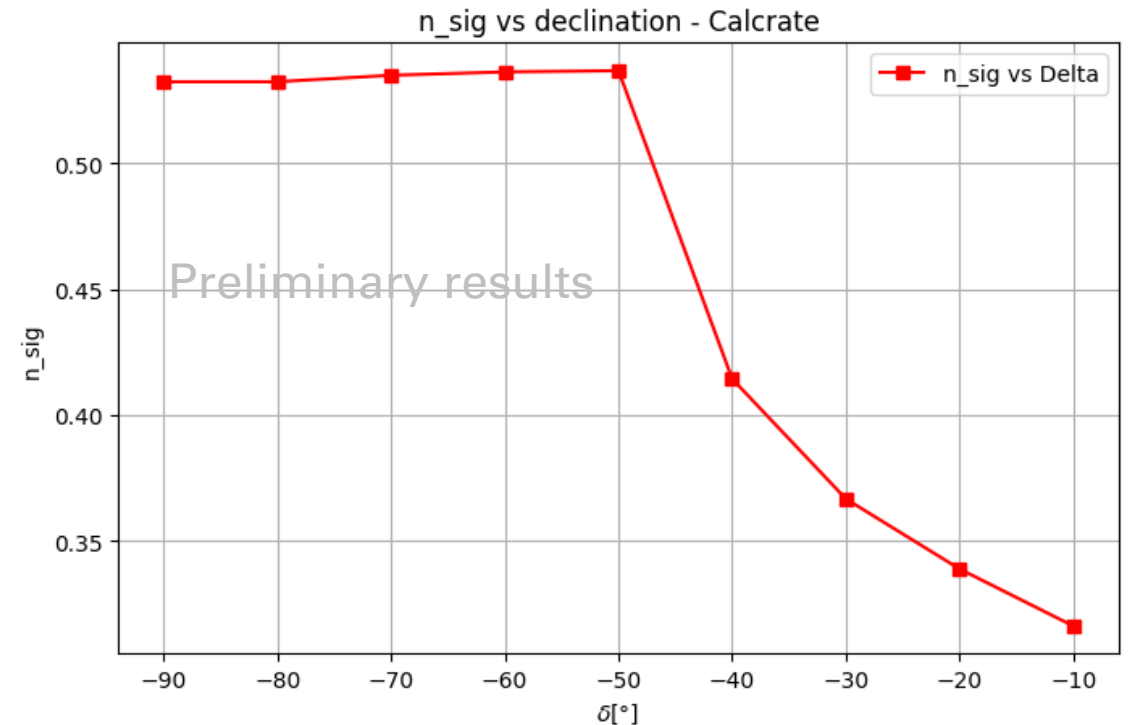
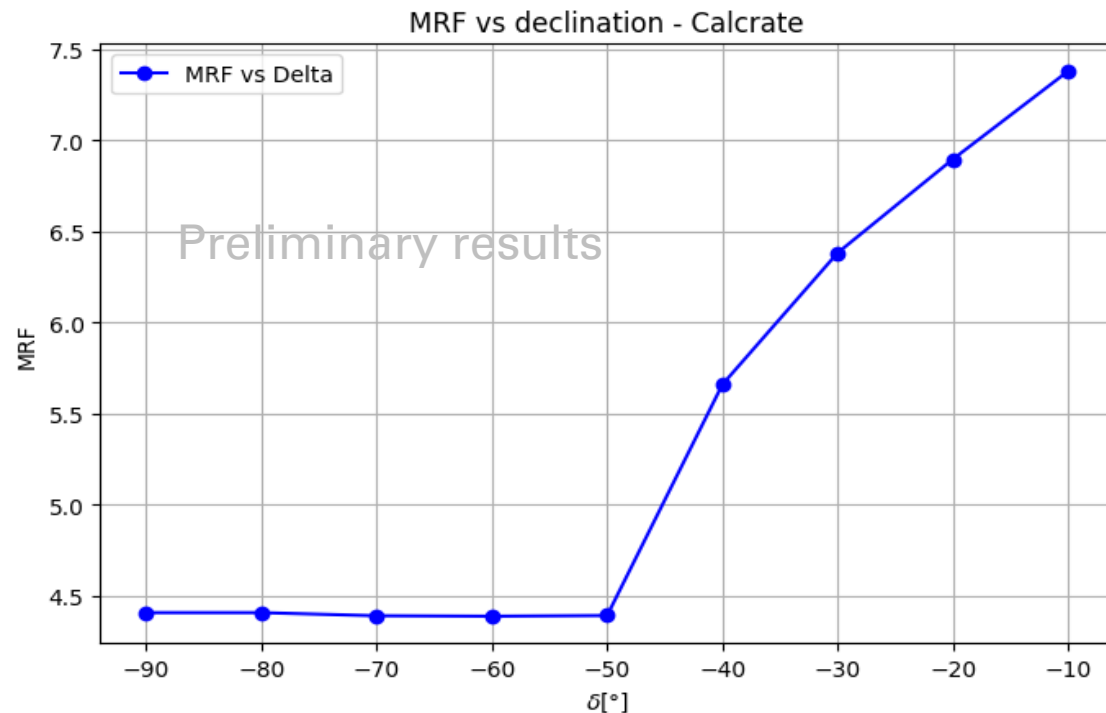
MRF – Calcrate ($\overline{\nu}_\mu$)

Detector configuration: ARCA21

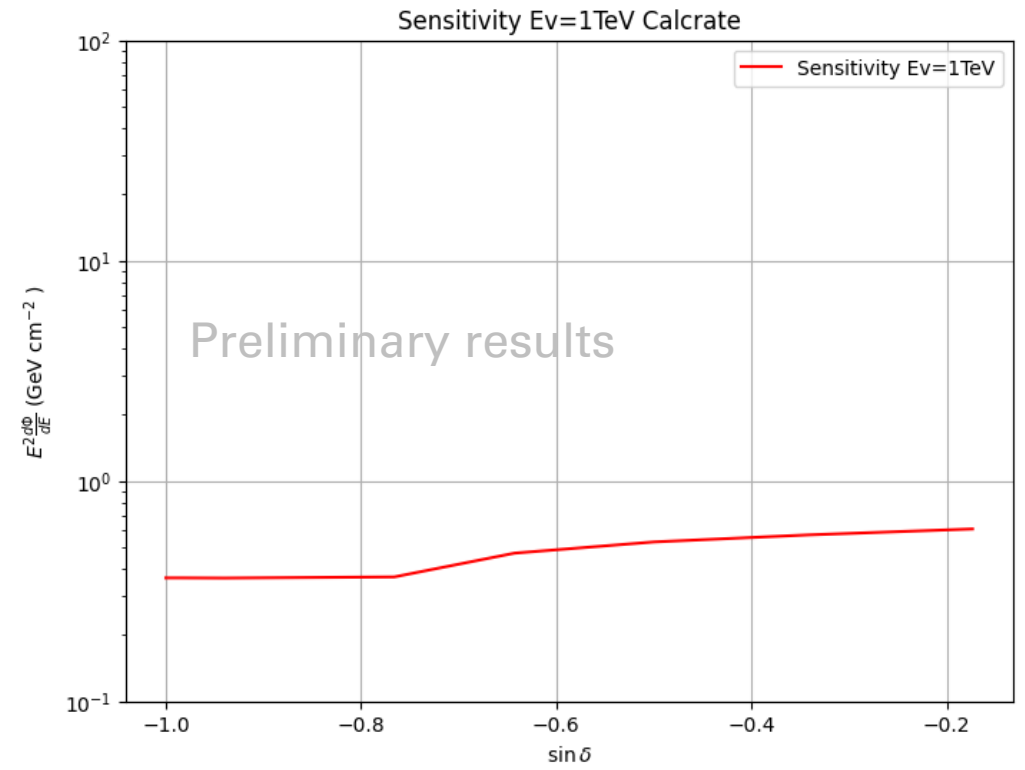
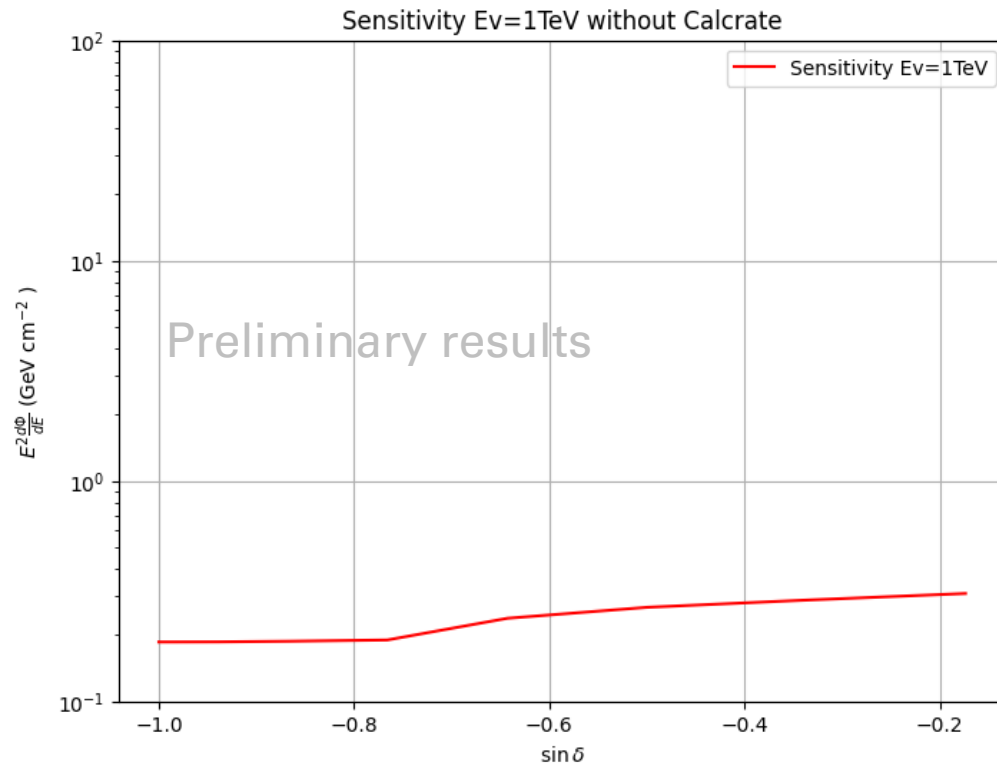
(202475_5_53_29_DETECTORESPONSE_ARCA21_133_tmva_tracks_v10.ALL_zen_b40_pre_upm01_antinoise_bdt095.root)

Source distance: 1Mpc

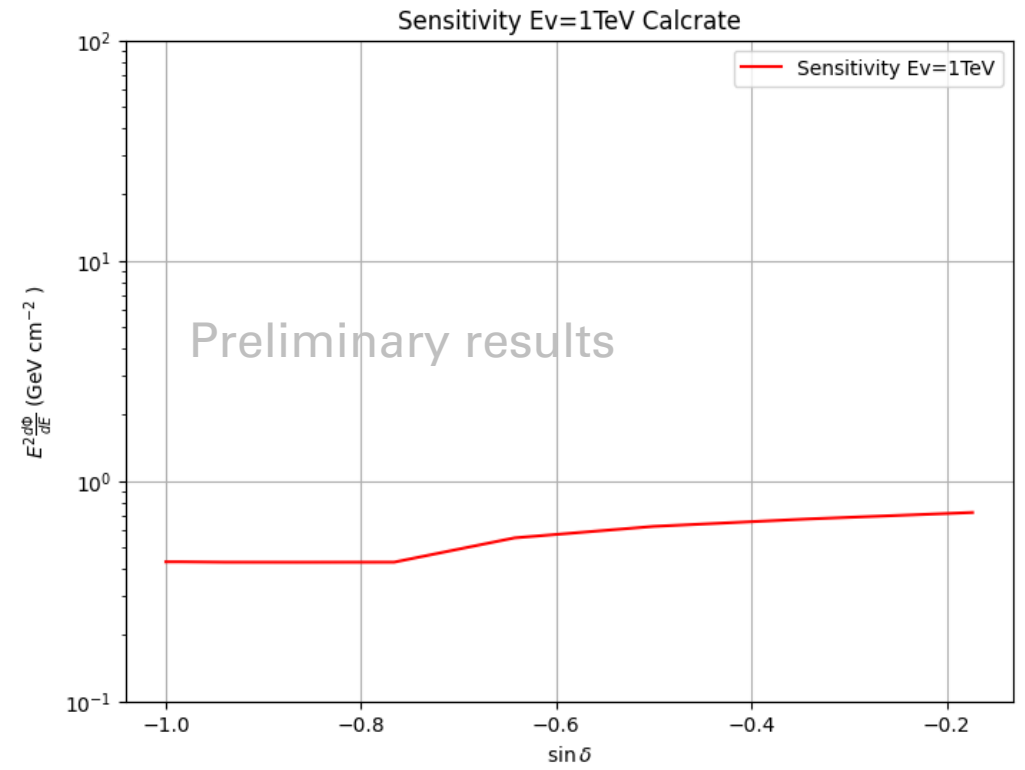
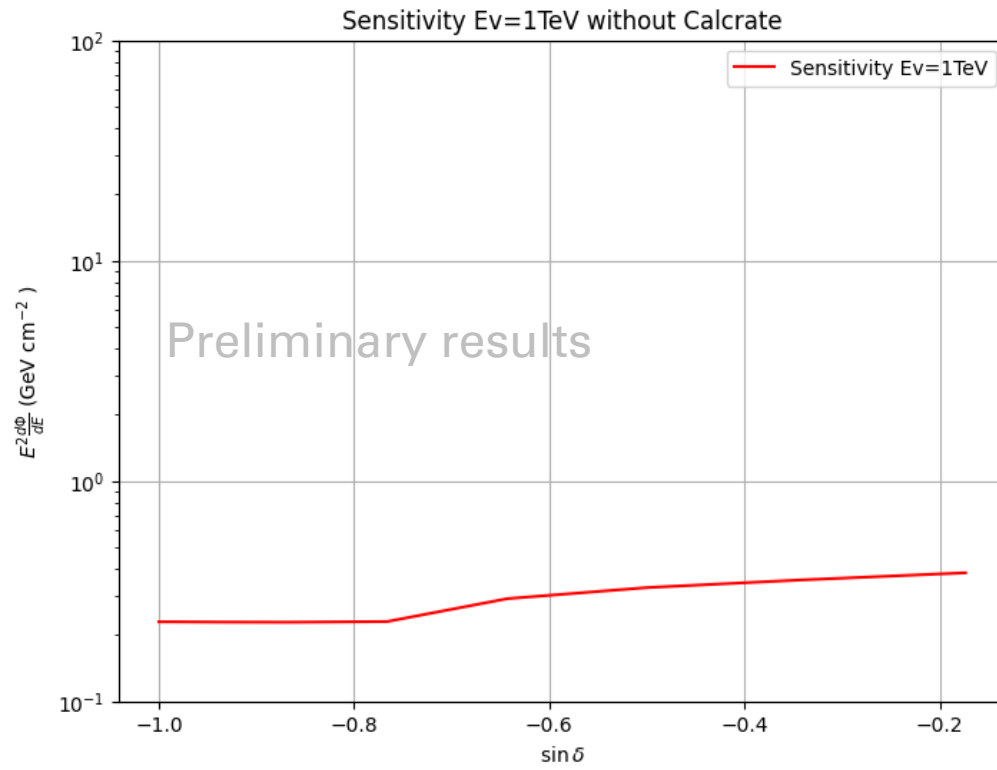
Supernova Model: SN IIn with CSM interactions



COMPARISON SENSITIVITY (ν_μ)



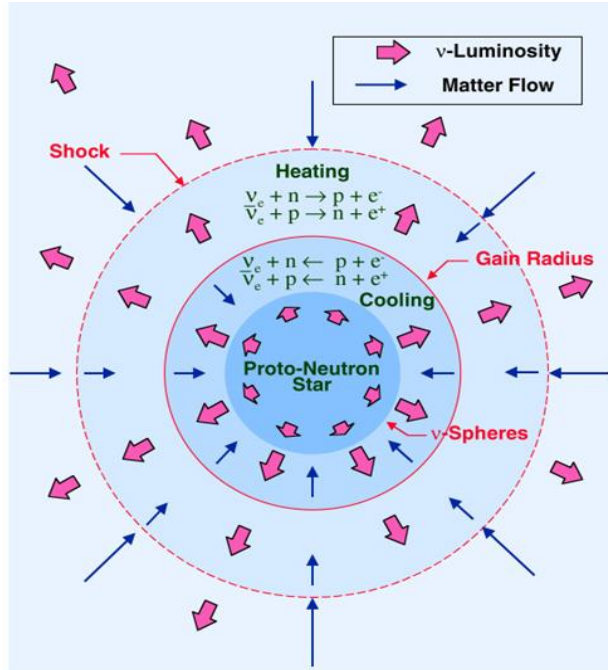
COMPARISON SENSITIVITY ($\bar{\nu}_\mu$)



BACK-UP



PHYSICAL SCENARIO



- The core-collapse supernovae (CCSNe) explosion mechanism is driven by low energy (MeV) neutrinos, which release most of the system's gravitational binding energy.
- In addition to MeV neutrinos, some CCSNe are good candidates for the production of high-energy (H-E) neutrinos (from GeV to PeV)

PHYSICAL SCENARIO

The potential to extend the detection horizon past the Magellanic Clouds through detecting HE neutrinos has to be explored.

- These neutrinos could give us also an insight into the cosmic ray acceleration processes, which are inaccessible with the low-energy neutrinos produced through nuclear processes in supernovae.

We consider two production mechanisms for H-E neutrinos in CCSNe:

interaction between SN ejecta and CSM
([Murase 2018](#))

relativistic choked jets
([Enberg et al. 2009](#))

EMISSION MODEL

Model

The model used for our studies is the one described in the Murase's works ([1], [2])

The neutrino energy fluence is described as follows:

SNe Candidates

- CSM interaction candidates: ***SN type IIn and II-P***
- Relativistic CJs candidates: ***SN type Ib/c and IIb***

$$E_\nu^2 \phi_\nu \sim \frac{1}{4\pi d^2} \frac{1}{2} \underbrace{\min[1, f_{pp}]}_{\text{pp optical depth}} \underbrace{\frac{\epsilon_{\text{cr}} \mathcal{E}_s}{\mathcal{R}_{\text{cr}10}}}_{\text{Spectrum dependent factor}} \underbrace{\left(\frac{E_\nu}{0.4 \text{ GeV}} \right)^{2-s}}_{\text{Proton momentum index}}$$

Fraction of energy carried by CR

Spectrum dependent factor

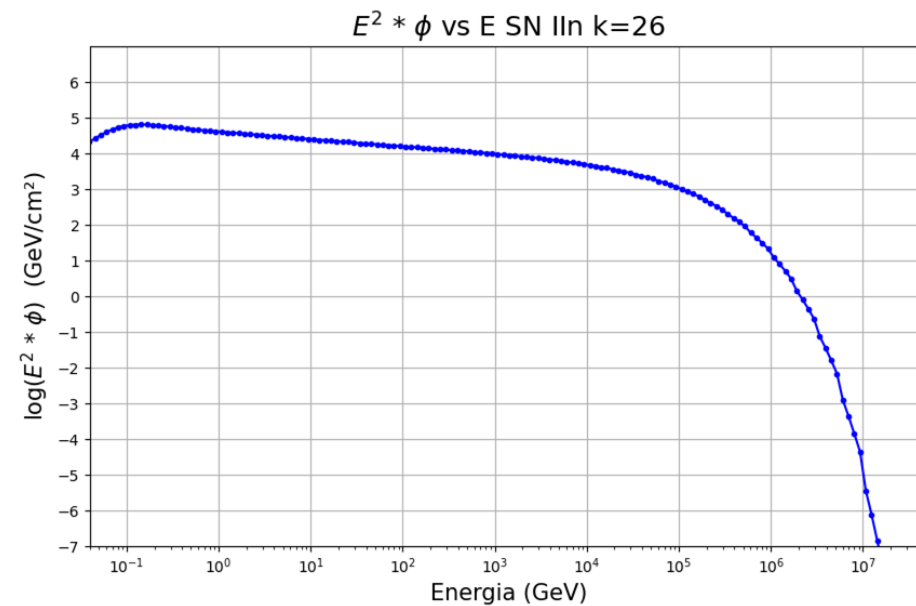
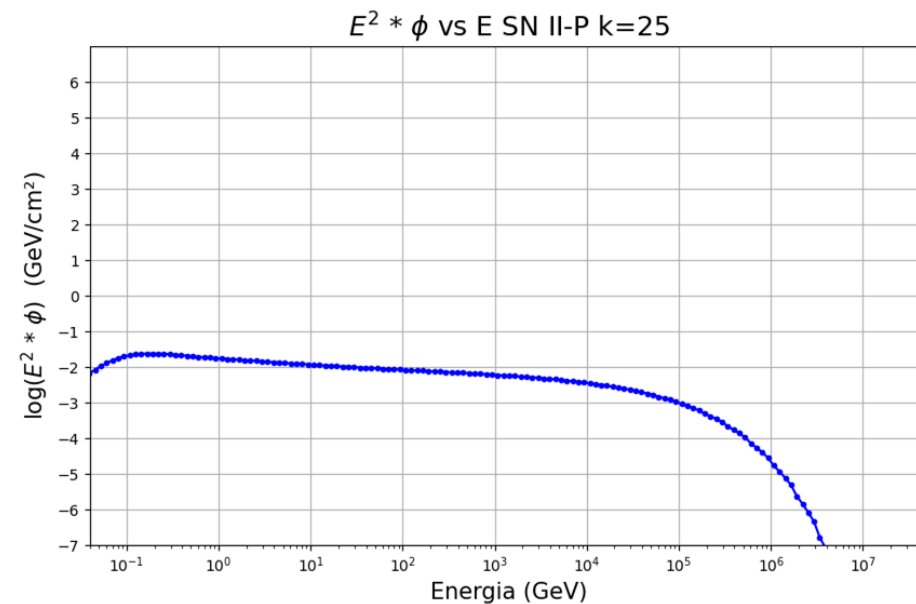
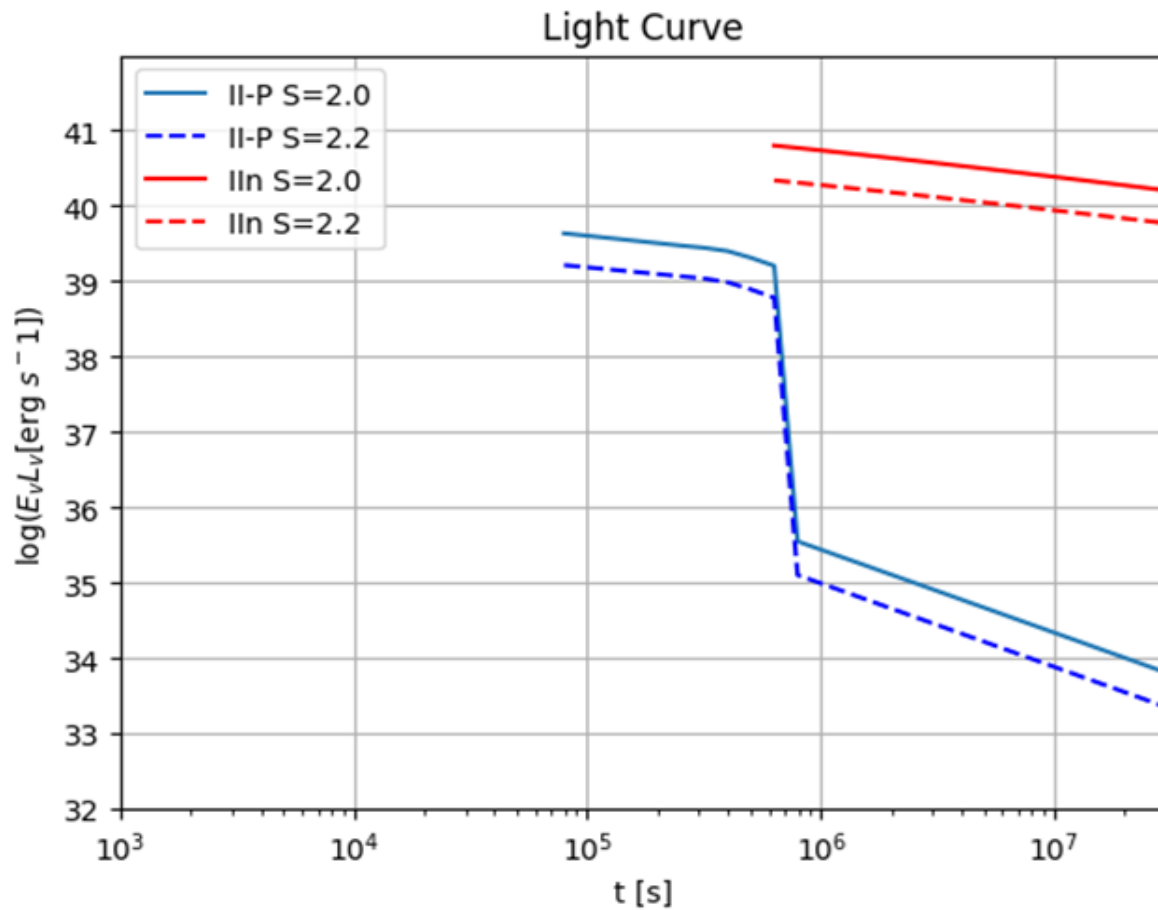
pp optical depth

Spectrum dependent factor

Proton momentum index

We started focusing on H-E neutrinos from CSM interaction production mechanism.

EMISSION MODEL



SOURCES CATALOG

Producing the catalogue consists of

- selecting SNe Type II (IIn or IIP);
- choosing the sources according to the life time of ARCA (from ARCA6 to ARCA21);
- selecting the declinations in the range from -90° to 40° (according to KM3NeT field of view)

SN_NAME	SN_TYPE	RA	DEC	Extended Source	Galactic SN	red_shift	distance (Mpc)	Alert data
SN2024 gyt	II		140	0.9	0 No	0.018	76.0	20/4/2024 7:42
SN2024ggi	II		170	-32.8	0 No	0.002435	10.3	11/4/2024 3:22
SN2024 gfc	II		78	-68.4	0 No	0.012	50.7	4/4/2024 8:08
SN2024 fpb	II		287	-17.9	0 No	0.022132	93.5	29/3/2024 11:19
SN2024 fpf	II		215	22.0	0 No	0.0152	64.2	4/4/2024 12:40
SN2024 faq	II		236	40.4	0 No	0.026	109.8	28/3/2024 10:40
SN2024exw	II		183	14.8	0 No	0.007	29.6	24/3/2024 16:11
SN2024evl	II		171	0.6	0 No	0.026364	111.3	22/3/2024 6:02
SN2024elf	II		264	40.0	0 No	0.03	126.7	15/3/2024 13:42
SN2024ehs	II		163	17.6	0 No	0.003776	15.9	15/3/2024 11:16
SN2024ees	II		188	14.5	0 No	0.0235	99.2	11/3/2024 6:44
SN2024eek	II		279	-62.5	0 No	0.015	63.3	11/3/2024 8:26
SN2024ecu	II		284	-42.4	0 No	0.02	84.4	9/3/2024 0:45
SN2024ecr	II		205	-16.6	0 No	0.052	219.6	10/3/2024 0:13
SN2024dym	II		154	-27.4	0 No	0.014	59.1	5/3/2024 22:14
SN2021qqu	II		54	-15.0	0 No	0.002	8.4	20/6/2021 10:19
SN2019fcn	II		122	-28.1	0 No	0.003	12.7	8/5/2019 23:02

PLAN FOR THE ANALYSIS

Estimation of the **sensitivity** for different configurations of ARCA detector to H-E ν coming from SN explosion.

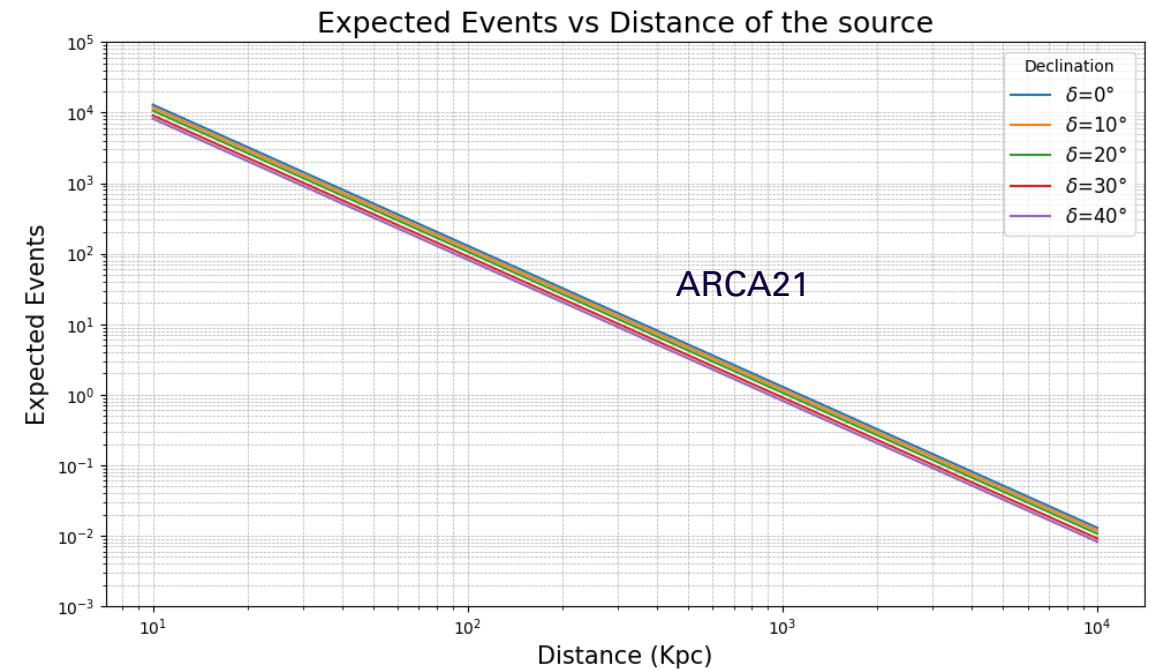
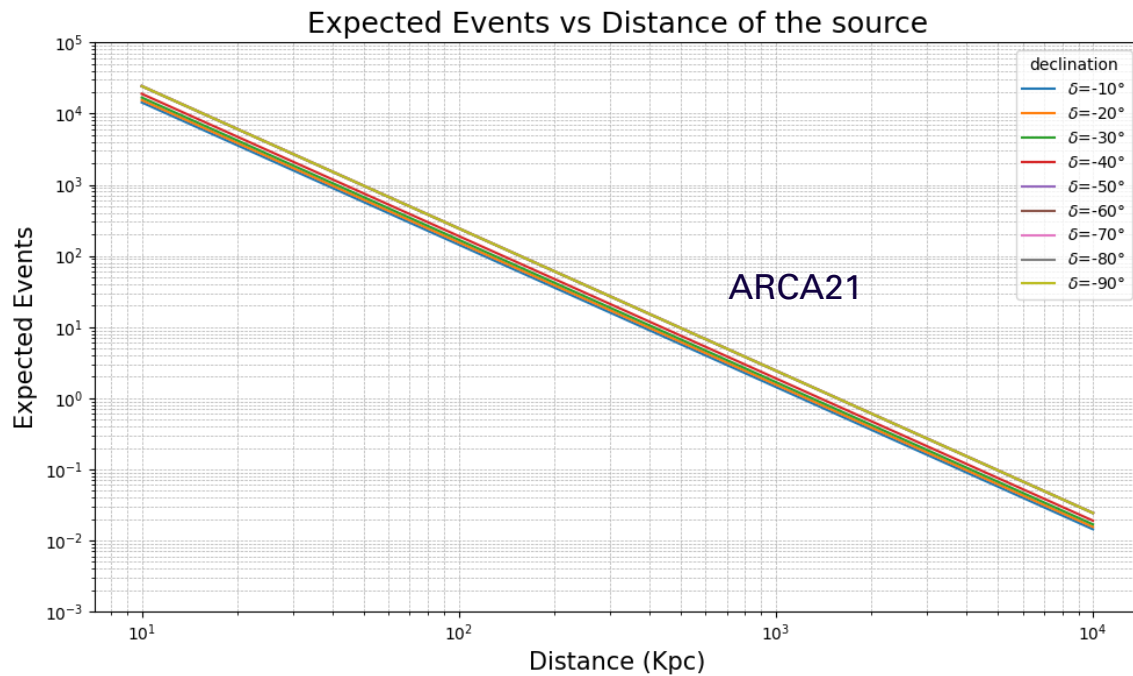
Two approaches to be compared:

1. Time dependent unbinned Likelihood using *Aanet*
2. Cut&Count analysis

Preliminary study: Evaluation of the expected number of neutrino events using the model and different configurations of ARCA

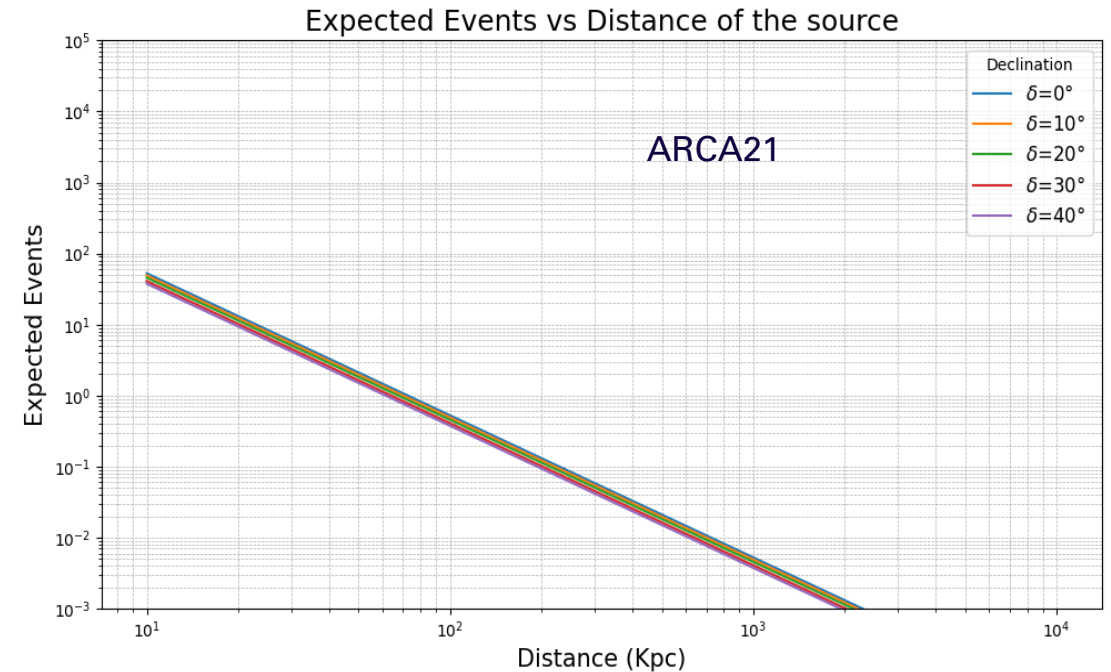
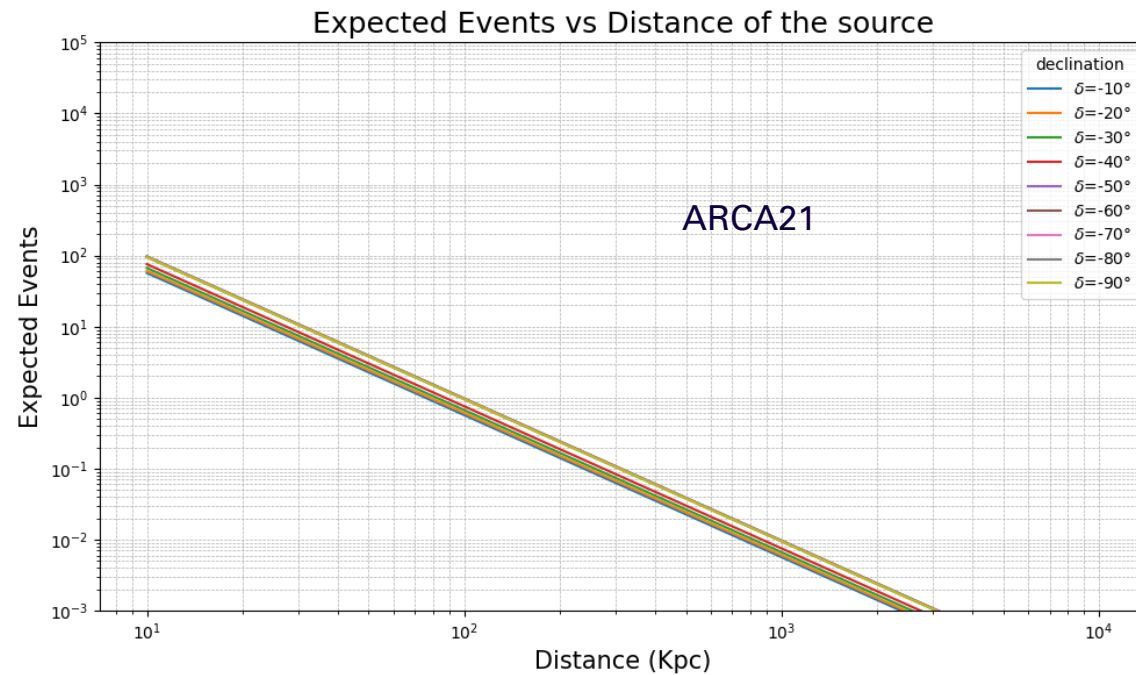
PLAN FOR THE ANALYSIS

We chose to plot the number of events vs. the source distance in a range from 10 Kpc to 10 Mpc, for each declination $\rightarrow$ *SN type II*



PLAN FOR THE ANALYSIS

We chose to plot the number of events vs. the source distance in a range from 10 Kpc to 10 Mpc, for each declination $\rightarrow$ *SN type IIP*



CUT&COUNT ANALYSIS

Using calcrate only for bkg (ν_μ)

summary fortrack channel.										
year	decl	cone	minlog	total-sig	total-bg	exp.limit	MRF	n	limit	discovery MDP
1	-10	1	1	0.790227	0.122613	2.37344	3.00349	2.6		3.29019
1	-20	1	1	0.822549	0.136043	2.38106	2.89474	2.6		3.16091
1	-30	1	1	0.884013	0.150242	2.3891	2.70256	2.6		2.94113
1	-40	1	1	1.02049	0.169901	2.40017	2.35199	2.6		2.54781
1	-50	1	1	1.44285	0.220479	2.42842	1.68307	3.5		2.42575
1	-60	1	1	1.30071	0.208647	2.42184	1.86193	3.5		2.69083
1	-70	1	1	1.25419	0.198654	2.41627	1.92656	3.5		2.79064
1	-80	1	1	1.19663	0.186288	2.40936	2.01346	2.5		2.0892
1	-90	1	1	1.19663	0.186288	2.40936	2.01346	2.5		2.0892

Using calcrate only for bkg ($\overline{\nu}_\mu$)

summary fortrack channel.										
year	decl	cone	minlog	total-sig	total-bg	exp.limit	MRF	n	limit	discovery MDP
1	-10	1	1	0.710384	0.0511665	2.33243	3.28333	2.7		3.80076
1	-20	1	1	0.740897	0.0573695	2.33602	3.15296	2.7		3.64423
1	-30	1	1	0.797263	0.0637286	2.33969	2.93466	2.7		3.38659
1	-40	1	1	0.92157	0.0723304	2.34465	2.54419	2.7		2.92978
1	-50	1	1	1.30571	0.0948036	2.35756	1.80558	2.6		1.99125
1	-60	1	1	1.16342	0.0873281	2.35328	2.02273	2.6		2.2348
1	-70	1	1	1.11724	0.079557	2.34881	2.10234	2.6		2.32717
1	-80	1	1	1.06349	0.0756675	2.34658	2.20649	2.6		2.44478
1	-90	1	1	1.06349	0.0756675	2.34658	2.20649	2.6		2.44478

My code output for n_sig e MRF

```
Calculating MRF for Anumu:
n_bk: 0.0511665, n_sig: 0.6322, AUL: 2.4830120463640655, MRF: 3.92757362601086
n_bk: 0.0573695, n_sig: 0.6831, AUL: 2.489829590389203, MRF: 3.644897658306548
n_bk: 0.0637286, n_sig: 0.7419, AUL: 2.496631317929238, MRF: 3.3651857634846176
n_bk: 0.0723084, n_sig: 0.8364, AUL: 2.503234038470522, MRF: 2.9928670952540912
n_bk: 0.0948036, n_sig: 1.07, AUL: 2.5254267785886038, MRF: 2.3602119426061714
n_bk: 0.0853128, n_sig: 1.074, AUL: 2.517504113933246, MRF: 2.344044798820527
n_bk: 0.079557, n_sig: 1.07, AUL: 2.5116206021416487, MRF: 2.3473089739641577
n_bk: 0.0756675, n_sig: 1.066, AUL: 2.5092457146956835, MRF: 2.3538890381760633
n_bk: 0.0756675, n_sig: 1.066, AUL: 2.5092457146956835, MRF: 2.3538890381760633

Calculating MRF for Numu:
n_bk: 0.122613, n_sig: 0.8053, AUL: 2.54776751241586, MRF: 3.163749549752713
n_bk: 0.136043, n_sig: 0.8684, AUL: 2.5585466122273965, MRF: 2.946276614725238
n_bk: 0.150242, n_sig: 0.942, AUL: 2.574513931237921, MRF: 2.73302965099567
n_bk: 0.169921, n_sig: 1.061, AUL: 2.5912245094335287, MRF: 2.4422474169967283
n_bk: 0.220479, n_sig: 1.355, AUL: 2.6349655705497823, MRF: 1.9446240373061123
n_bk: 0.208647, n_sig: 1.366, AUL: 2.6231042274508645, MRF: 1.9202812792466064
n_bk: 0.198653, n_sig: 1.37, AUL: 2.6145991341361823, MRF: 1.9084665212672862
n_bk: 0.186288, n_sig: 1.365, AUL: 2.6024341631178007, MRF: 1.9065451744452753
n_bk: 0.186288, n_sig: 1.365, AUL: 2.6024341631178007, MRF: 1.9065451744452753
```

CUT&COUNT ANALYSIS

Passing my flux to Calcrate2.py with GraphFlux()

ν_{μ}

summary fortrack channel.										
year	decl	cone	minloge	total-sig	total-bg	exp.limit	MRF	n	limit	discovery MDP
1	-10	1	1	0.382147	0.122613	2.37344	6.21081	2.6		6.80367
1	-20	1	1	0.407962	0.136043	2.38106	5.83649	2.6		6.37315
1	-30	1	1	0.440563	0.150242	2.3891	5.42283	2.6		5.90154
1	-40	1	1	0.497599	0.169901	2.40017	4.82351	2.6		5.2251
1	-50	1	1	0.645541	0.220479	2.42842	3.76183	3.5		5.42181
1	-60	1	1	0.648057	0.208647	2.42184	3.73708	3.5		5.40076
1	-70	1	1	0.649841	0.198654	2.41627	3.71826	3.5		5.38593
1	-80	1	1	0.646868	0.186288	2.40936	3.72466	2.5		3.86478
1	-90	1	1	0.646868	0.186288	2.40936	3.72466	2.5		3.86478

$\overline{\nu_{\mu}}$

summary fortrack channel.										
year	decl	cone	minloge	total-sig	total-bg	exp.limit	MRF	n	limit	discovery MDP
1	-10	1	1	0.31614	0.0511665	2.33243	7.37783	2.7		8.54052
1	-20	1	1	0.338821	0.0573695	2.33602	6.89455	2.7		7.96882
1	-30	1	1	0.366676	0.0637286	2.33969	6.38082	2.7		7.36345
1	-40	1	1	0.41437	0.0723304	2.34465	5.65836	2.7		6.51592
1	-50	1	1	0.536904	0.0948036	2.35756	4.39103	2.6		4.84258
1	-60	1	1	0.536398	0.0873281	2.35328	4.38719	2.6		4.84715
1	-70	1	1	0.535064	0.079557	2.34881	4.38978	2.6		4.85923
1	-80	1	1	0.532494	0.0756675	2.34658	4.40676	2.6		4.88268
1	-90	1	1	0.532494	0.0756675	2.34658	4.40676	2.6		4.88268

MRF – 10KPC SN IIN

